

Learning to Refer: How Scene Complexity Affects Emergent Communication in Neural Agents

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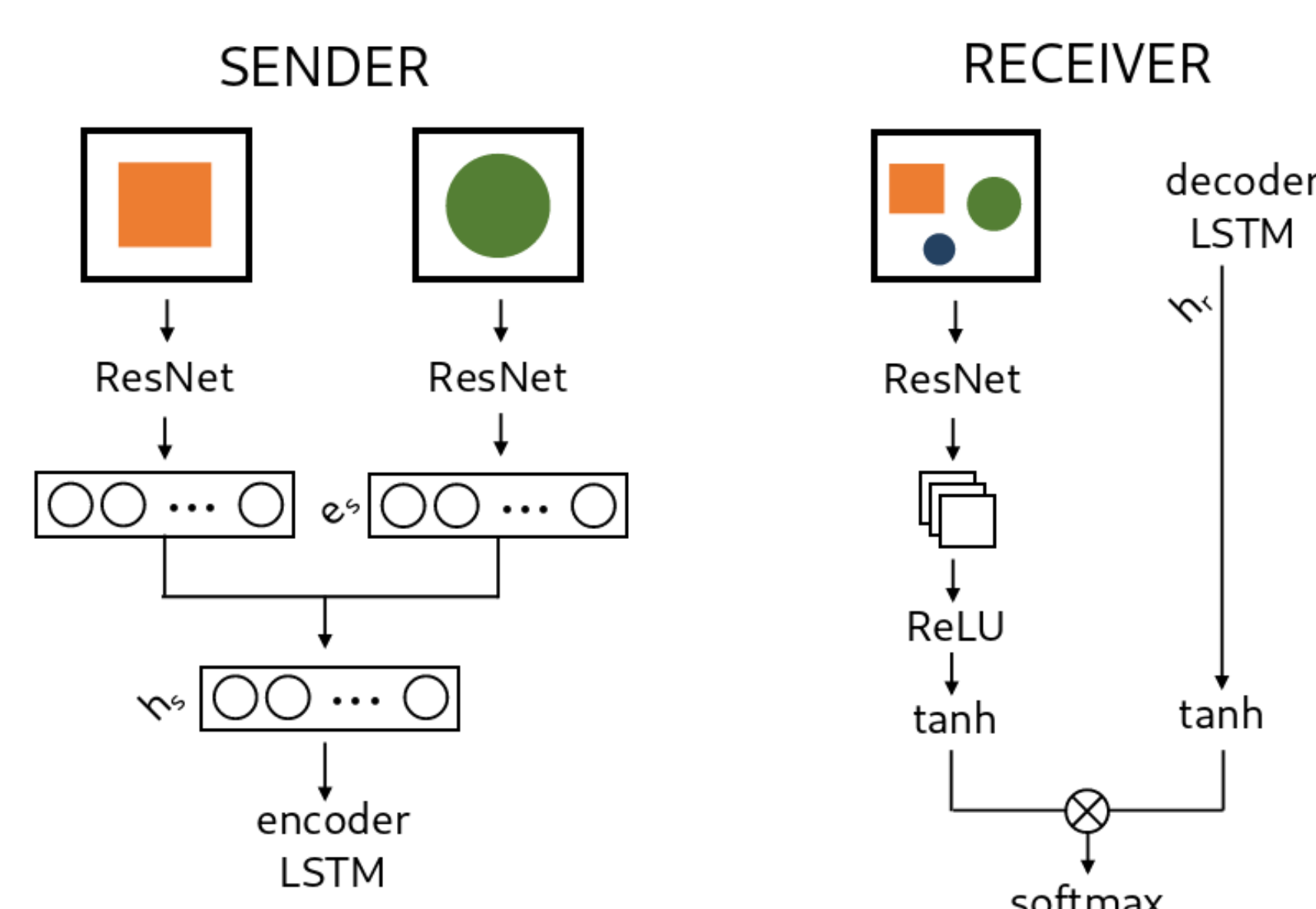
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Aims

A picture can say thousand words...

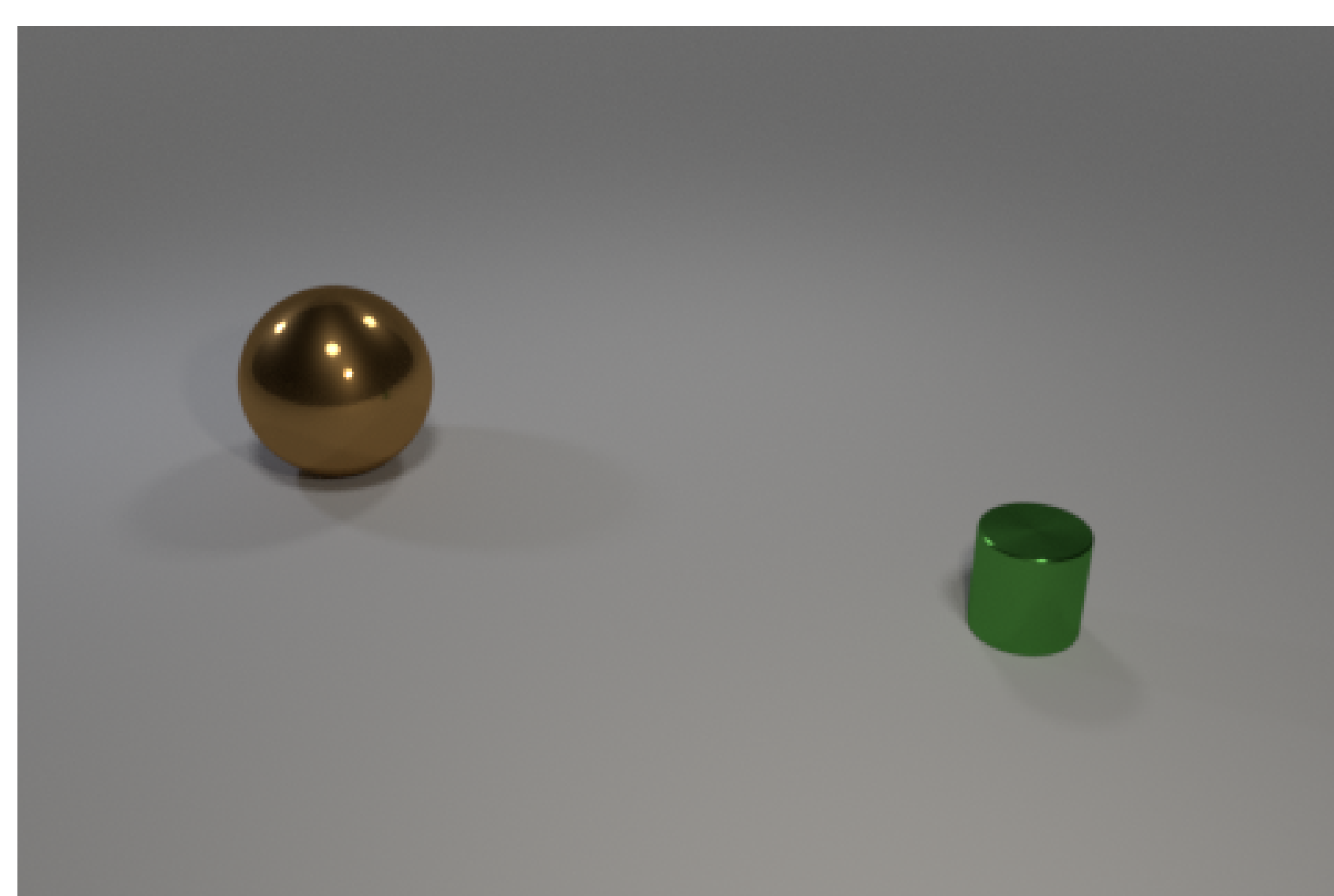
- Study **emergence of discrete linguistic symbols and languages** from continuous sensory input in **interactive communication**
- Study **constraints** on symbol grounding in interacting neural agents
- Investigate the **role of scene complexity, discriminating features present, and message space** on emergent languages

Architecture

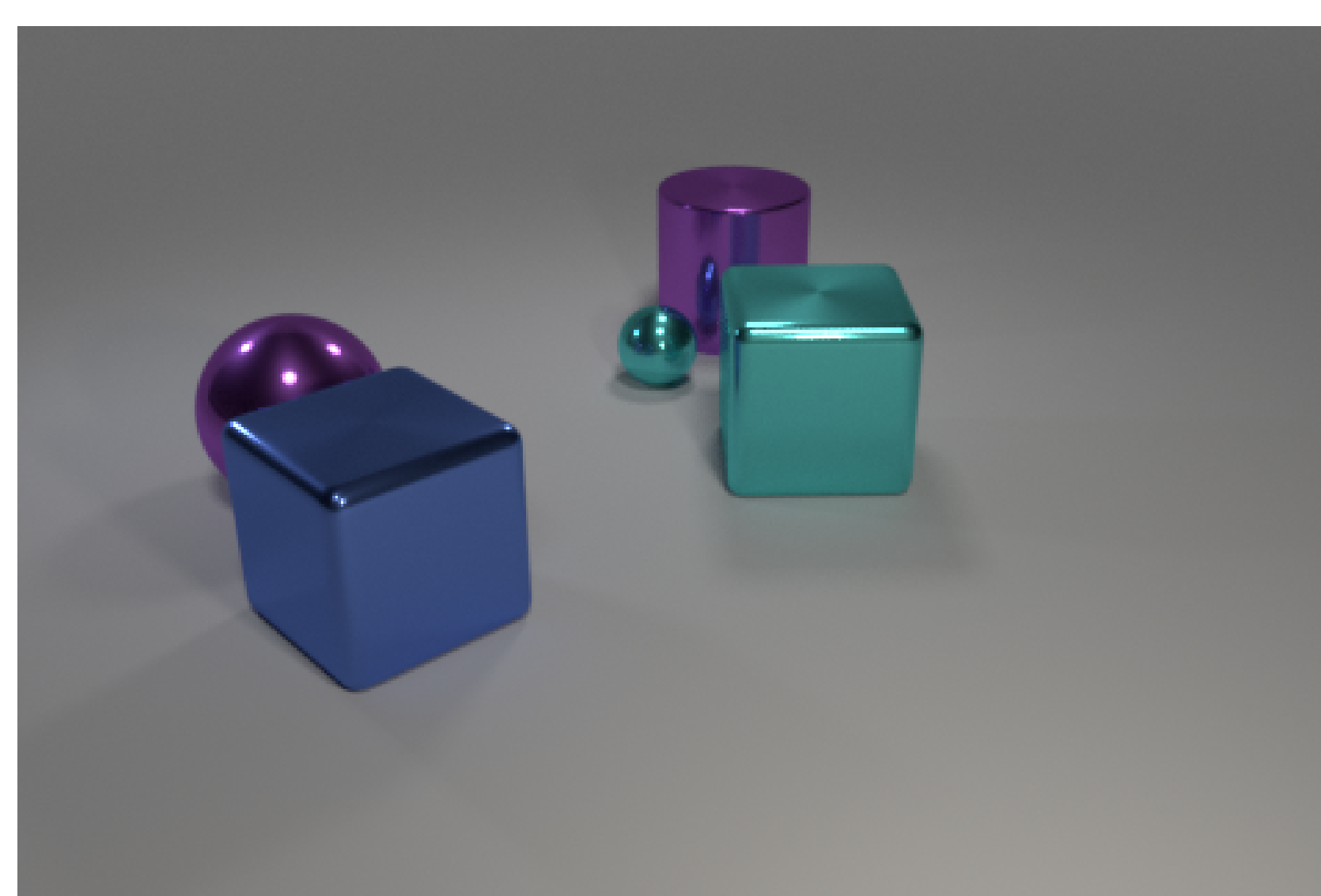


Conclusions

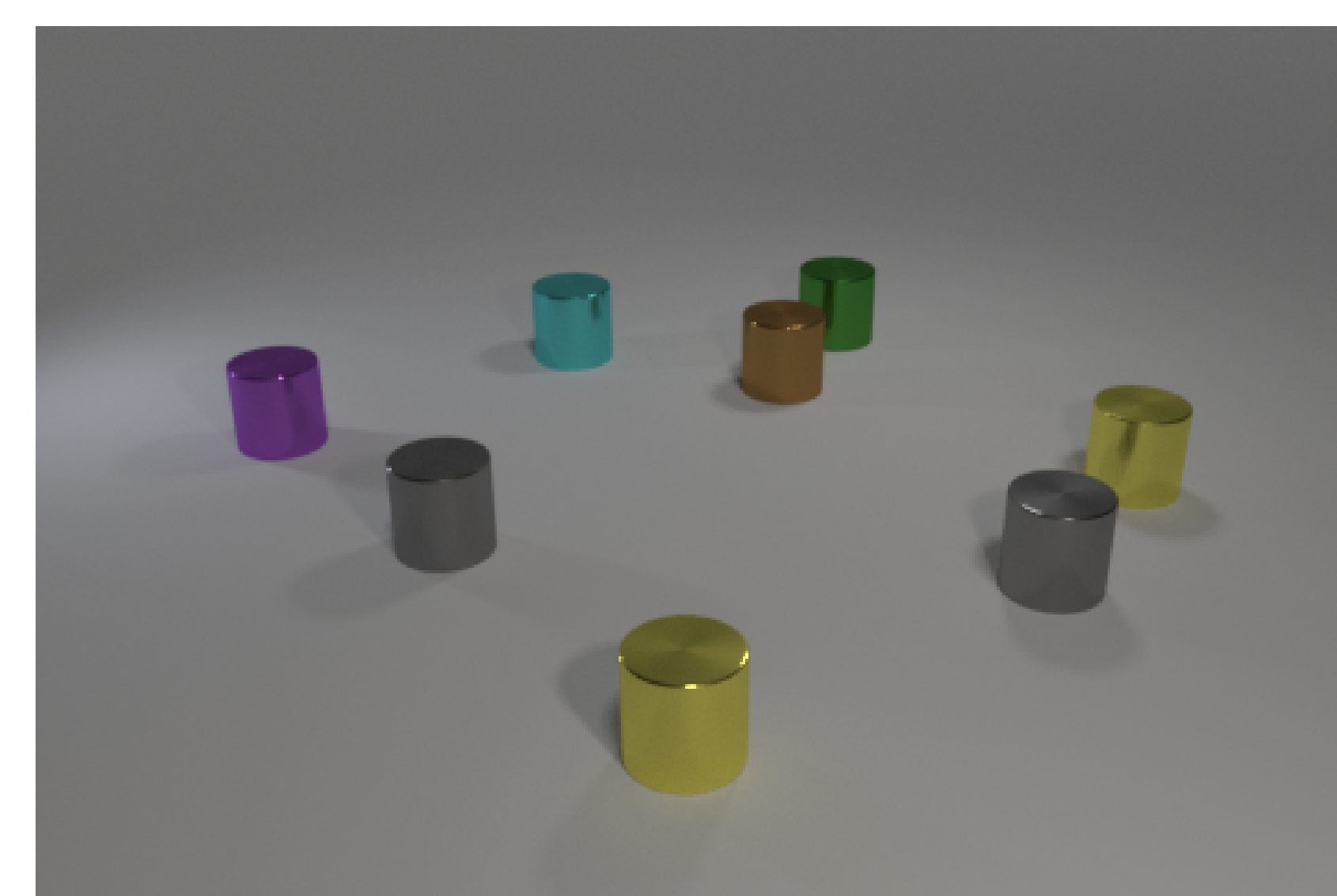
- Medium vocabularies ($|V| \in \{10, 16\}$) and message lengths ($n \in \{2, 4\}$) perform best
- Scene complexity (number of distractors, number of features) dramatically impacts communication success
- Shape attributes easiest to learn, size attributes most challenging
- Future: Analyze linguistic properties of emergent languages



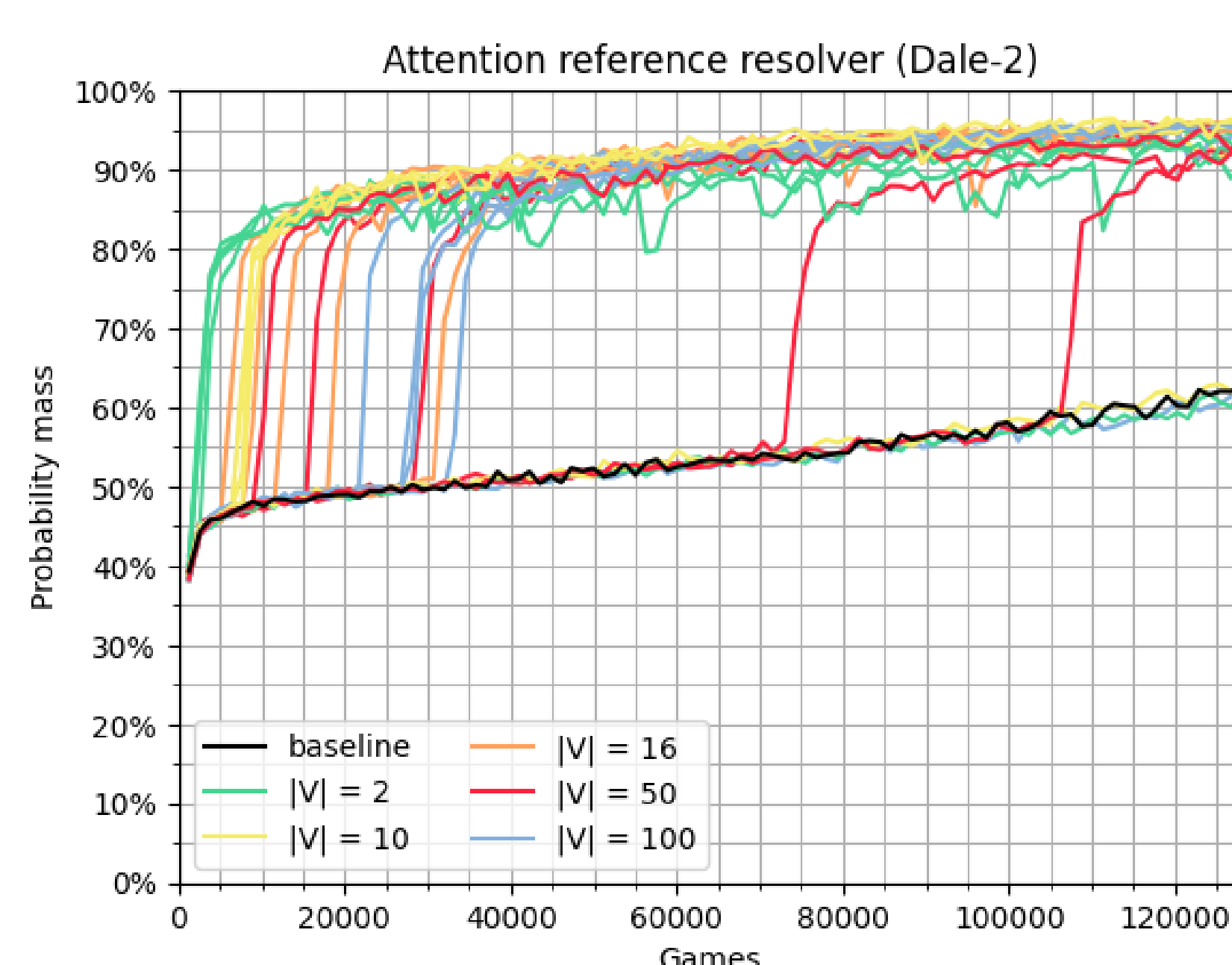
(a) 'Dale-2', small green cylinder



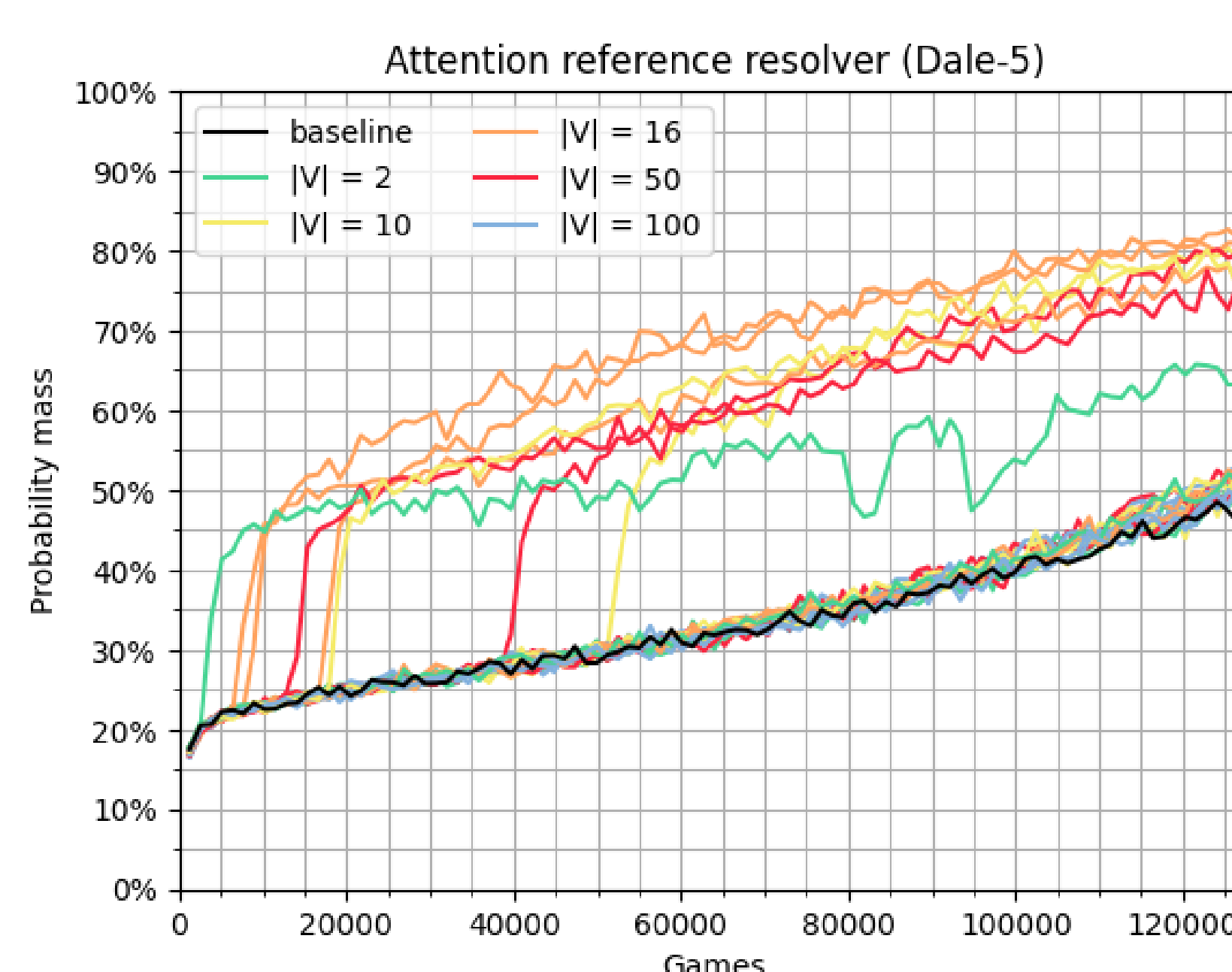
(b) 'Dale-5', large purple cylinder



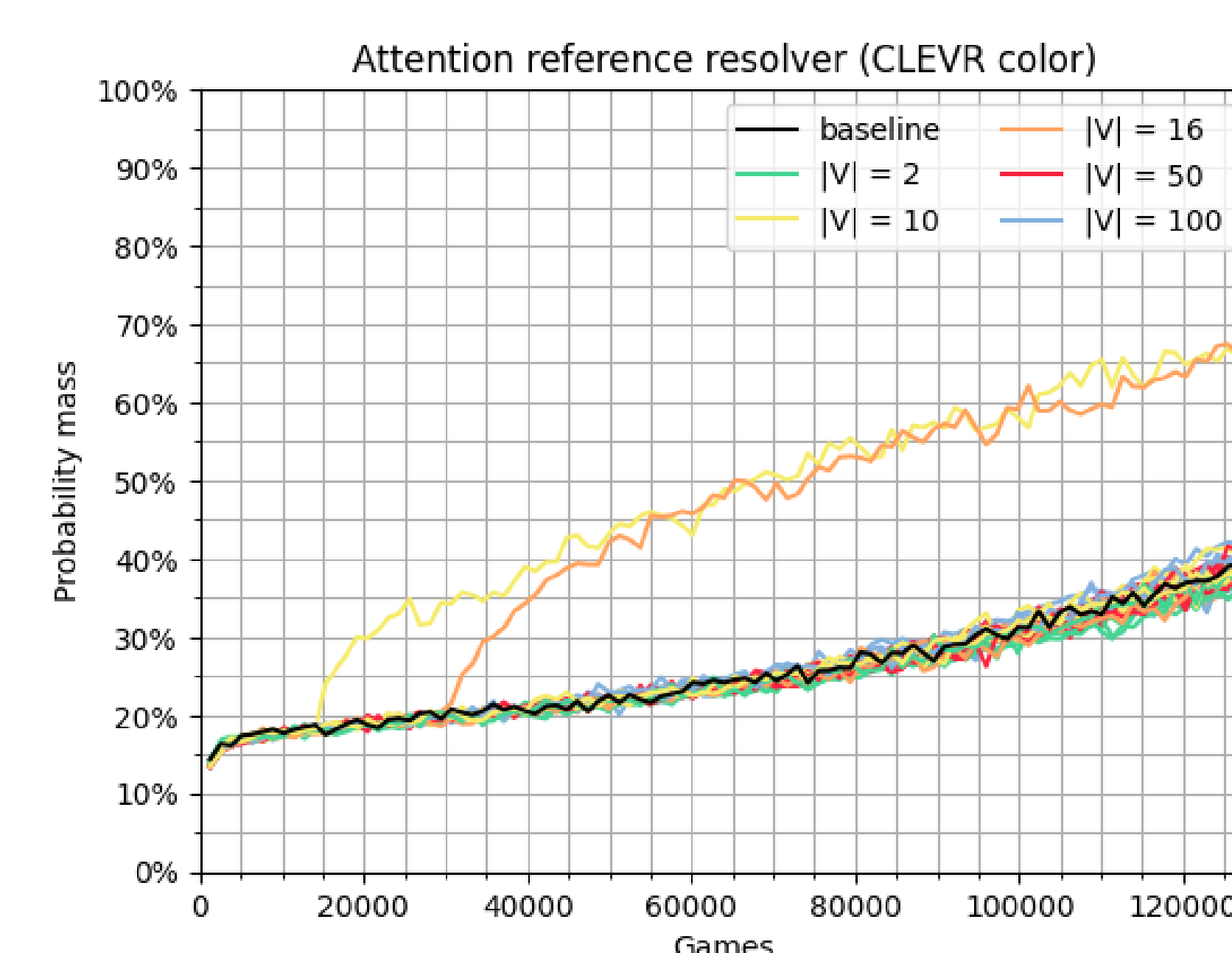
(c) 'CLEVR color', small brown cylinder



(a)



(b)



(c)

Dale-2 Dataset

- **Accuracy:** 99%
- **F1-Score:** 98.57%
- **Shape:** 99.75% precision
- **Color:** 98.09% precision
- **Size:** 98.73% precision

Dale-5 Dataset

- **Accuracy:** 69%
- **F1-Score:** 89.53%
- **Shape:** 98.30% precision
- **Color:** 92.44% precision
- **Size:** 69.43% precision

CLEVR Color Dataset

- **strict string accuracy:** 93%
- **avg F1-Score on attributes:** 95.17%
- **Shape:** 100% precision
- **Color:** 92.76% precision
- **Size:** N/A (not used)